

MOTION IN A PLANE

✚ Important formulas and concepts directly from the NCERT-

- A vector quantity has both a magnitude and a direction and obeys the triangle law of addition or parallelogram law of addition.
- The magnitude of displacement is either less or equal to the path length of an object b/w two points.
- For motion in two or three dimensions, velocity and acceleration vectors may have any angle b/w 0° to 180° .
- Motion in a plane (two dimensions) can be treated as two separate simultaneous one-dimensional motions with constant acceleration along two perpendicular directions.

❖ Projectile motion-

- An object that is in flight after being thrown or projected is called a **projectile**.
- This motion is a result of two separate components of motion.
- One component is along a horizontal direction without any acceleration and the other along the vertical direction with constant acceleration due to the force of gravity.
- The components of initial velocity,

Horizontal, $v_x = v \cos \theta$
 Vertical, $v_y = v \sin \theta$
- The components of velocity at time t ,

Horizontal, $v_x = v \cos \theta$
 Vertical, $v_y = v \sin \theta - gt$
- Positions at time t ,

Horizontal, $x = (v_x)t = (v \cos \theta)t$
 Vertical, $y = (v_y)t = (v \sin \theta)t - \frac{1}{2}gt^2$

○ Equation of path of projectile-

➤ $y = x \tan \theta (1 - x/R)$

or $y = x \tan \theta - gx^2/2(v \cos \theta)^2$

➤ the eqⁿ is in the form of $ax+bx^2$, this is the eqⁿ of parabola so, the path of projectile is parabola.

➤ Path of one projectile w.r.t. other projectile,

1) If relative acceleration is zero, path is *straight line*.

2) If horizontal component same, path is *vertically straight*.

3) If vertical component same, path is *horizontally straight*.

○ time of maximum height-

➤ time to reach maximum height,

$$t = v \sin \theta / g$$

➤ total time of projectile,

$$t = 2v \sin \theta / g$$

this is k/as '*time of flight*'.

○ Maximum height of a projectile-

➤ $H_{\max} = (u \sin \theta)^2 / 2g$

○ Horizontal range of a projectile-

➤ The horizontal distance travelled by a projectile is k/as 'horizontal range'.

➤ $R = v^2 \sin 2\theta / g$

➤ R is maximum when $\theta = 45^\circ$ which is,

$$R_{\max} = v^2 / g$$

➤ For two projectiles, if speed of projection is same and angle of projection are θ and $(90-\theta)$ then, range is same.

❖ River boat problems-

➤ **Case 1 – cross river along shortest possible path or reach just opposite end (velocity of boat makes an angle with displacement{width of river})**

- v_R = speed of river
 v_b = speed of boat

$$v_R = v_b \cos \theta \quad (\theta = \text{angle made by boat from horizontal})$$

- time of crossing, $t = \text{displacement} / v_b \sin \theta$

➤ **Case 2 – to cross the river in minimum time (velocity of boat is in the direction of displacement)**

- In this case we observe a drift D ,

$$D = v_R (\text{displacement} / v_b)$$
- Time of crossing, $t = \text{displacement} / v_b$

❖ Rain man problems-

➤ **Case 1 – already raining in vertical downward direction and observation in oblique direction.**

- Velocity of rain w.r.t man,

$${}_M v_R = v_R - v_M$$
- Man has to hold umbrella at,

$$\tan \theta = v_M / v_R$$

➤ **Case 2 – already raining in oblique direction and observation in vertical downward direction.**

- $v_M = v_R \sin \theta$ (θ = angle made by rain with vertical)
- velocity of rain w.r.t. man,

$${}_M v_R = v_R \cos \theta$$

❖ **Uniform circular motion(UCM)-**

- **Angular velocity, $\omega = d\theta/dt$ (θ = angular displacement)**
- **Angular acceleration, $\alpha = d\omega/dt$**
- **Linear velocity, $v = R\omega$**
Tangential acceleration, $a_t = R\alpha$
- **Centripetal acceleration, $a_c = v^2/R = \omega^2 R = v\omega$**
- **In UCM, speed is constant.**
- **Angular speed is also constant.**
- **But velocity not constant (direction is changing continuously)**
- **Angular acceleration is zero (angular velocity is constant).**
- **Tangential acceleration is also constant (speed is constant and tangential acceleration is responsible for changing magnitude (speed) only).**
- **Centripetal acceleration is present.(direction towards centre always)**

SUMMARY

1. *Scalar quantities* are quantities with magnitudes only. Examples are distance, speed, mass and temperature.
2. *Vector quantities* are quantities with magnitude and direction both. Examples are displacement, velocity and acceleration. They obey special rules of vector algebra.
3. A vector **A** multiplied by a real number λ is also a vector, whose magnitude is λ times the magnitude of the vector **A** and whose direction is the same or opposite depending upon whether λ is positive or negative.
4. Two vectors **A** and **B** may be *added graphically* using *head-to-tail method* or *parallelogram method*.
5. Vector addition is *commutative* :

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

It also obeys the *associative law* :

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$$

6. A *null* or *zero vector* is a vector with zero magnitude. Since the magnitude is zero, we don't have to specify its direction. It has the properties :

$$\mathbf{A} + \mathbf{0} = \mathbf{A}$$

$$\lambda \mathbf{0} = \mathbf{0}$$

$$0 \mathbf{A} = \mathbf{0}$$

7. The *subtraction* of vector **B** from **A** is defined as the sum of **A** and **-B** :

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

8. A vector **A** can be *resolved* into component along two given vectors **a** and **b** lying in the same plane :

$$\mathbf{A} = \lambda \mathbf{a} + \mu \mathbf{b}$$

where λ and μ are real numbers.

9. A *unit vector* associated with a vector **A** has magnitude 1 and is along the vector **A**:

$$\hat{\mathbf{n}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

The unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are vectors of unit magnitude and point in the direction of the *x*-, *y*-, and *z*-axes, respectively in a right-handed coordinate system.

10. A vector **A** can be expressed as

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}}$$

where A_x, A_y are its components along *x*-, and *y* -axes. If vector **A** makes an angle θ

with the *x*-axis, then $A_x = A \cos \theta$, $A_y = A \sin \theta$ and $A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2}$, $\tan \theta = \frac{A_y}{A_x}$.

11. Vectors can be conveniently added using *analytical method*. If sum of two vectors **A** and **B**, that lie in *x-y* plane, is **R**, then :

$$\mathbf{R} = R_x \hat{\mathbf{i}} + R_y \hat{\mathbf{j}}, \text{ where, } R_x = A_x + B_x, \text{ and } R_y = A_y + B_y$$

12. The *position vector* of an object in *x-y* plane is given by $\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}}$ and the *displacement* from position **r** to position **r'** is given by

$$\Delta \mathbf{r} = \mathbf{r}' - \mathbf{r}$$

$$= (x' - x) \hat{\mathbf{i}} + (y' - y) \hat{\mathbf{j}}$$

$$= \Delta x \hat{\mathbf{i}} + \Delta y \hat{\mathbf{j}}$$

13. If an object undergoes a displacement $\Delta \mathbf{r}$ in time Δt , its *average velocity* is given by

$\mathbf{v} = \frac{\Delta \mathbf{r}}{\Delta t}$. The *velocity* of an object at time *t* is the limiting value of the average velocity as Δt tends to zero :

$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt}$. It can be written in unit vector notation as :

$$\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}} \quad \text{where} \quad v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$$

When position of an object is plotted on a coordinate system, $\mathbf{v}$ is always tangent to the curve representing the path of the object.

14. If the velocity of an object changes from $\mathbf{v}$ to $\mathbf{v}'$ in time Δt , then its *average acceleration*

is given by: $\bar{\mathbf{a}} = \frac{\mathbf{v} - \mathbf{v}'}{\Delta t} = \frac{\Delta \mathbf{v}}{\Delta t}$

The *acceleration* $\mathbf{a}$ at any time t is the limiting value of $\bar{\mathbf{a}}$ as $\Delta t \rightarrow 0$:

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt}$$

In component form, we have : $\mathbf{a} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} + a_z \hat{\mathbf{k}}$

where, $a_x = \frac{dv_x}{dt}$, $a_y = \frac{dv_y}{dt}$, $a_z = \frac{dv_z}{dt}$

15. If an object is moving in a plane with constant acceleration $a = |\mathbf{a}| = \sqrt{a_x^2 + a_y^2}$ and its position vector at time $t = 0$ is $\mathbf{r}_0$, then at any other time t , it will be at a point given by:

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$$

and its velocity is given by :

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a} t$$

where $\mathbf{v}_0$ is the velocity at time $t = 0$

In component form :

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$$

$$v_x = v_{0x} + a_x t$$

$$v_y = v_{0y} + a_y t$$

Motion in a plane can be treated as superposition of two separate simultaneous one-dimensional motions along two perpendicular directions

16. An object that is in flight after being projected is called a *projectile*. If an object is projected with initial velocity $\mathbf{v}_0$ making an angle θ_0 with x -axis and if we assume its initial position to coincide with the origin of the coordinate system, then the position and velocity of the projectile at time t are given by :

$$x = (v_0 \cos \theta_0) t$$

$$y = (v_0 \sin \theta_0) t - (1/2) g t^2$$

$$v_x = v_{0x} = v_0 \cos \theta_0$$

$$v_y = v_0 \sin \theta_0 - g t$$

The path of a projectile is *parabolic* and is given by :

$$y = (\tan \theta_0) x - \frac{g x^2}{2 (v_0 \cos \theta_0)^2}$$

The *maximum height* that a projectile attains is :

$$h_m = \frac{(v_o \sin \theta_o)^2}{2g}$$

The time taken to reach this height is :

$$t_m = \frac{v_o \sin \theta_o}{g}$$

The horizontal distance travelled by a projectile from its initial position to the position it passes $y = 0$ during its fall is called the *range*, R of the projectile. It is :

$$R = \frac{v_o^2}{g} \sin 2\theta_o$$

17. When an object follows a circular path at constant speed, the motion of the object is called *uniform circular motion*. The magnitude of its acceleration is $a_c = v^2/R$. The direction of a_c is always towards the centre of the circle. The angular speed ω , is the rate of change of angular distance. It is related to velocity v by $v = \omega R$. The acceleration is $a_c = \omega^2 R$. If T is the time period of revolution of the object in circular motion and ν is its frequency, we have $\omega = 2\pi \nu$, $v = 2\pi \nu R$, $a_c = 4\pi^2 \nu^2 R$

POINTS TO PONDER

1. The path length traversed by an object between two points is, in general, not the same as the magnitude of displacement. The displacement depends only on the end points; the path length (as the name implies) depends on the actual path. The two quantities are equal only if the object does not change its direction during the course of motion. In all other cases, the path length is greater than the magnitude of displacement.
2. In view of point 1 above, the average speed of an object is greater than or equal to the magnitude of the average velocity over a given time interval. The two are equal only if the path length is equal to the magnitude of displacement.
3. The vector equations (3.33a) and (3.34a) do not involve any choice of axes. Of course, you can always resolve them along any two independent axes.
4. The kinematic equations for uniform acceleration do not apply to the case of uniform circular motion since in this case the magnitude of acceleration is constant but its direction is changing.
5. An object subjected to two velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ has a resultant velocity $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. Take care to distinguish it from velocity of object 1 relative to velocity of object 2 : $\mathbf{v}_{12} = \mathbf{v}_1 - \mathbf{v}_2$. Here $\mathbf{v}_1$ and $\mathbf{v}_2$ are velocities with reference to some common reference frame.
6. The resultant acceleration of an object in circular motion is towards the centre only if the speed is constant.
7. The shape of the trajectory of the motion of an object is not determined by the acceleration alone but also depends on the initial conditions of motion (initial position and initial velocity). For example, the trajectory of an object moving under the same acceleration due to gravity can be a straight line or a parabola depending on the initial conditions.